WEEKLY LESSON PLAN · 2026-2027

Principles of Applied Engineering

TEKS §127.781 · CTE Level 1 · Grade 9 · Mr. Gavaldon · Da Vinci School

Week 1

MONDAY Day 1 - Procedures, Syllabus & Your Design Notebook

Aug 10

OBJECTIVE	Students will set up a standards-compliant engineering design notebook (cover, table of contents, numbered and dated pages) and write a complete first entry with a labeled sketch and measurements recorded with correct units and tolerance, then evaluate a sample entry against a professional-documentation rubric.
TEKS	§127.781(d)(1) professional standards & employability; (d)(3)(B) maintain a design & computation engineering notebook
ELPS / EB SUPPORT	Model every notebook step live under the doc cam with icons and sentence stems (VC, CD); pre-teach notebook, entry, tolerance, and witness with picture cards (PV, DP). Give sentence frames for the integrity scenario ('The professional response is ___ because ___') and extra time on the units table (RS, ET). A bilingual peer is available for Google Classroom sign-in as backup, but the visual model carries it (PN).
AGENDA · 45 MIN	**7’** Bell-work: You invent something valuable; a competitor says they thought of it first. What evidence would prove it was you? (write in notebook) **10’** Why engineers document: IP/legal, reproducibility, traceability, quality & safety (notes + notebook standards) **8’** Anatomy of an entry + units/tolerance, modeled live under the doc cam **12’** Set up notebook (cover, TOC, page numbers) + write a complete first entry with a labeled, dimensioned sketch **5’** Rubric self-audit + Google Classroom sign-in **3’** Exit ticket: 3 features that make a notebook legally defensible
SLIDES / ACTIVITY	The Engineer's Notebook: Your Professional Record
NOTEBOOK	Set it up today: cover, Table of Contents (pgs 1-3), number pages, date entries. First entry = what you hope to build this year.
TURN IN	Engineering Notebook - Setup, First Entry & Documentation Audit

TUESDAY Day 2 - What Is Engineering? (video + intro)

Aug 11

OBJECTIVE

Students will distinguish engineering from science, identify at least four engineering disciplines with a historical milestone, map an engineered product's subsystems to the disciplines and career paths behind them, and produce an annotated technical sketch (isometric or orthographic) with labeled subsystems and dimensioned callouts.

TEKS

§127.781(d)(2)(A) engineering disciplines & their history; (d)(2)(E) engineering career paths; (d)(3)(C) sketching to present ideas

ELPS / EB SUPPORT

Guided-viewing sheet gives sentence stems and the video is paused at each discipline (RS, CD, WT). Pre-teach discipline, constraint, orthographic, and isometric with picture cards (PV, DP). Pair-share the science-vs-engineering question before whole-class share so EB students rehearse (PN, WT). A word bank supports the careers table (VC).

AGENDA · 45 MIN

- 6′ Bell-work: list 5 engineered objects within reach; next to each, guess which kind of engineer made it
- 8′ Science vs engineering + design under constraints (criteria/constraints/trade-offs)
- 12′ Watch: Crash Course Engineering #1 + guided viewing (disciplines & history)
- 8′ History of the disciplines + careers, pay, and pathways
- 8′ Technical sketching mini-lesson + annotate an engineered object
- 3′ Exit ticket

SLIDES / ACTIVITY

What Engineers Do: Disciplines, History & the Sketch That Communicates It

NOTEBOOK

Glue in the guided-viewing sheet. Sketch + label one engineered object; name its branch and the problem it solves.

TURN IN

Engineering Fields, Careers & Technical Sketch

WEDNESDAY Day 3 - Paper Airplane Challenge (Design Process)

Aug 12

OBJECTIVE

Students will apply the full engineering design process to a constrained paper-aircraft challenge - defining criteria and constraints, selecting a concept with a weighted decision matrix, controlling variables for a fair test, running three trials, and computing average distance and percent improvement to justify an iteration with data.

TEKS

§127.781(d)(2)(G) the engineering design process; (d)(1)(B) collaborate as a team

ELPS / EB SUPPORT

The design-report sheet is the scaffold, released one part at a time (CD, VC). Pre-teach criteria, constraint, variable, and iteration with gestures and quick examples (PV). Mixed-level teams of 2-3 with defined jobs so EB students use process vocabulary in context (PN). A sentence frame is provided for the justification (RS).

AGENDA · 45 MIN

- 5′ Bell-work: what variables affect how far a paper plane flies? List 5 and circle the ONE you will change
- 8′ Watch TED-Ed + map its cycle onto the engineering design process
- 10′ Design depth: criteria vs constraints, controlled variables, fair testing, % improvement (worked)
- 17′ Build -> test (3 trials) -> analyze -> iterate -> retest (3 trials)
- 5′ Compute averages & percent change + exit ticket

SLIDES / ACTIVITY

Design Like an Engineer: The Paper-Aircraft Trials

NOTEBOOK

Glue in the design-process record sheet. Sketch your two candidate designs before building.

TURN IN

Paper-Aircraft Design Report (Fair Test + Decision Matrix)

THURSDAY — Day 4 - Paper Tower Challenge (iterate)

Aug 13

OBJECTIVE
Students will iterate a free-standing paper tower to maximize height-per-sheet efficiency, apply structural concepts (compression, buckling, triangulation, base ratio, center of gravity) to diagnose a failure mode, and use Build-1 data to justify a single-variable redesign, then quantify the improvement.

TEKS
§127.781(d)(2)(G) engineering design process — iteration & constraints; (d)(1)(B) teamwork

ELPS / EB SUPPORT
Reuse Day 3's process vocabulary plus load, compression, buckling, and triangulation with a quick physical demo (repetition builds language: RS, PV, DP). Roles (structural lead / builder / tester-recorder) give every student a clear job (CD). The efficiency and percent-improvement formulas are shown worked before students compute (VC, ET).

AGENDA · 45 MIN

- **5′** Bell-work: why is a triangle stronger than a square? Which fails first under load - and how?
- **6′** Go over it + assign roles (structural lead / builder / tester-recorder)
- **9′** Structural concepts: compression/tension, buckling, triangulation, base ratio, CG + the efficiency metric
- **18′** Build 1 -> test (height + 10 s stand) -> diagnose failure -> Build 2 -> retest
- **7′** Compute efficiency & percent improvement + reflection

SLIDES / ACTIVITY
Structures & Iteration: Build It Taller, Prove It's Better

NOTEBOOK
Log Build 1 and Build 2 heights. Note the one change you made and why. Sketch the strongest shape you found.

TURN IN
Structural Tower - Iteration & Efficiency Report

FRIDAY — Day 5 - Types of Engineering + Quiz Game

Aug 14

OBJECTIVE
Students will classify products by engineering discipline, analyze a technology's benefits, trade-offs, and unintended consequences on society, evaluate it with a weighted impact matrix, and defend an adopt/redesign/restrict recommendation with evidence.

TEKS
§127.781(d)(2)(A) fields of engineering; (d)(5) impact of technology on society

ELPS / EB SUPPORT
The impact matrix uses a word bank and numeric ratings so language load stays low while thinking stays high (VC, CD). Pre-teach discipline names, sustainability, trade-off, and unintended consequence with images (PV, DP). Team scoring lets EB students reason with a peer before writing (PN, WT). A sentence frame supports the recommendation (RS).

AGENDA · 45 MIN

- **5′** Bell-work: match 4 products to their engineering discipline and justify one
- **10′** Branches deep-dive + emerging/interdisciplinary fields + systems thinking
- **8′** Impact of technology: benefits, trade-offs, unintended consequences, life-cycle, ethics
- **12′** Team analysis: score a technology on a weighted impact matrix + defend a recommendation
- **7′** Spot-the-engineering audit (room/campus) with discipline + trade-off
- **3′** Exit ticket

SLIDES / ACTIVITY
Branches of Engineering & Their Impact on Society

NOTEBOOK
Record your 3 campus finds: what it is, its branch, and the problem it solves.

TURN IN
Technology Impact Analysis + Discipline Audit